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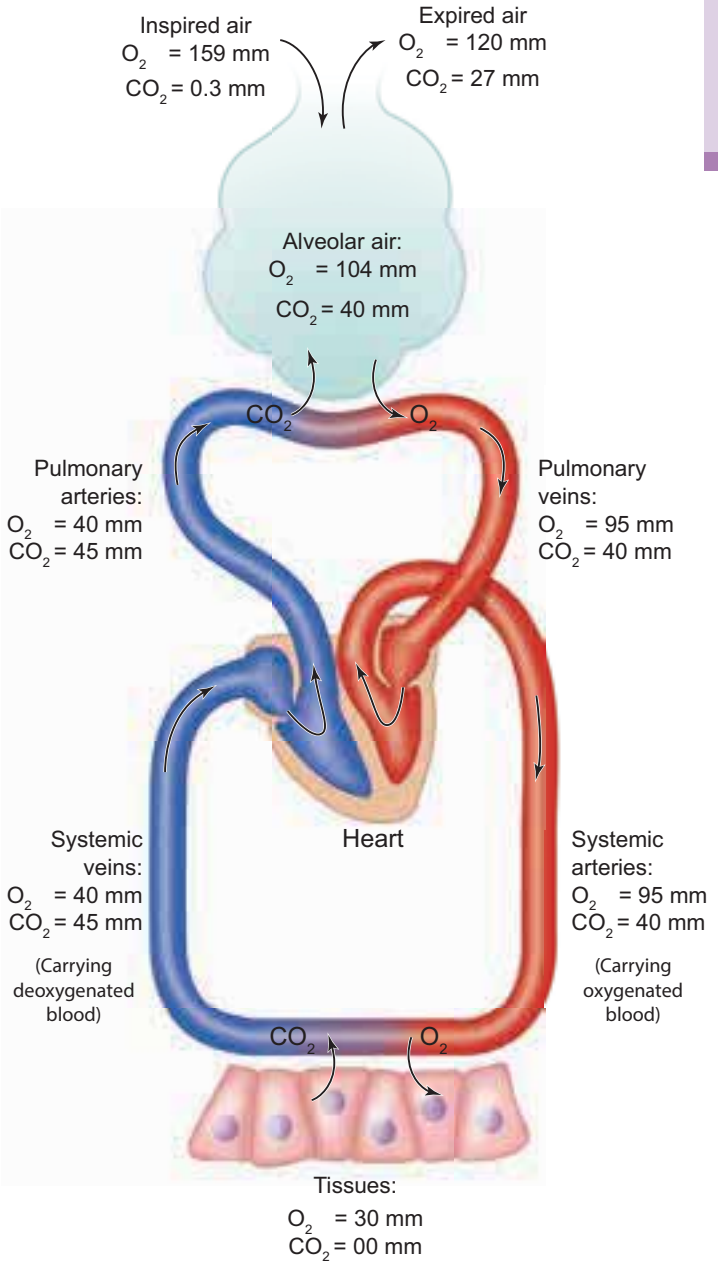


Figure 6.6 Exchange of gases at the alveolus and the tissue with blood and transport of oxygen and carbondioxide

Breathing through nose is healthier than through mouth– Why?

dioxide. Due to pressure gradients, O_2 from the alveoli enters into the blood and reaches the tissues. CO_2 enters into the blood from the tissues and reaches alveoli for elimination. As the solubility of CO_2 is 20–25 times higher than that of O_2, the partial pressure of CO_2 is much higher than that of O_2 (Tab.6.1 and Figure 6.6).

Respiratory pigments

Haemoglobin

Haemoglobin belongs to the class of conjugated protein. The iron containing pigment portion haem constitutes only 4% and the rest colourless protein of the histone class globin. Haemoglobin has a molecular weight of 68,000 and contains four atoms of iron, each of which can combine with a molecule of oxygen.

Methaemoglobin

If the iron component of the haem moieties is in the ferric state, than the normal ferrous state, it is called methaemoglobin. Methaemoglobin

Respiratory gases	Partial pressure mm Hg				
	Atmospheric air	Alveoli	Deoxygenated Blood	Oxygenated blood	Tissues
O_2	159	104	40	95	40
CO_2	0.3	40	45	40	45

Table 6.1 Partial pressure of Oxygen and Carbon dioxide (in mmHg) in comparison to those gases in the atmosphere